

Transconductance Amplifier

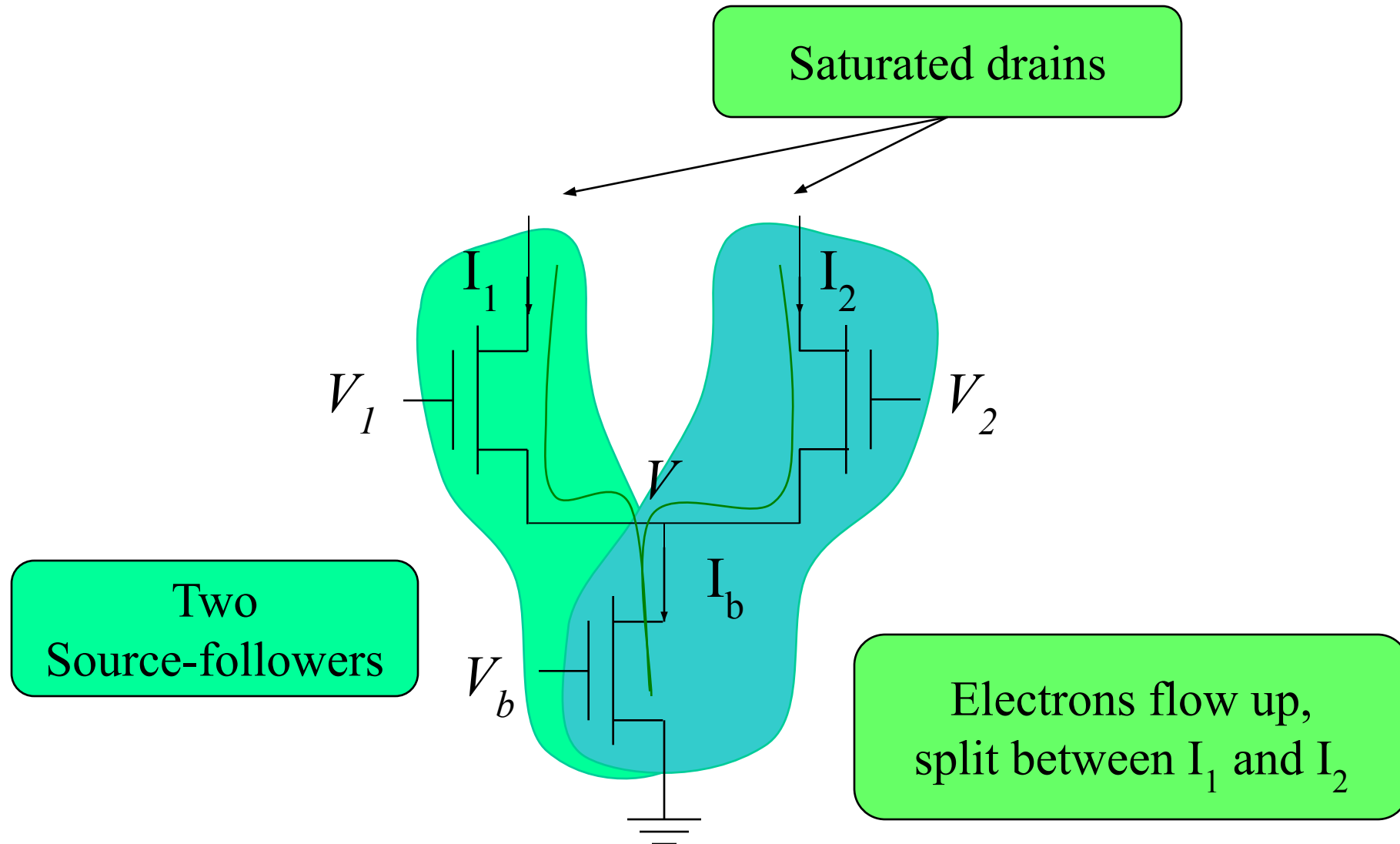
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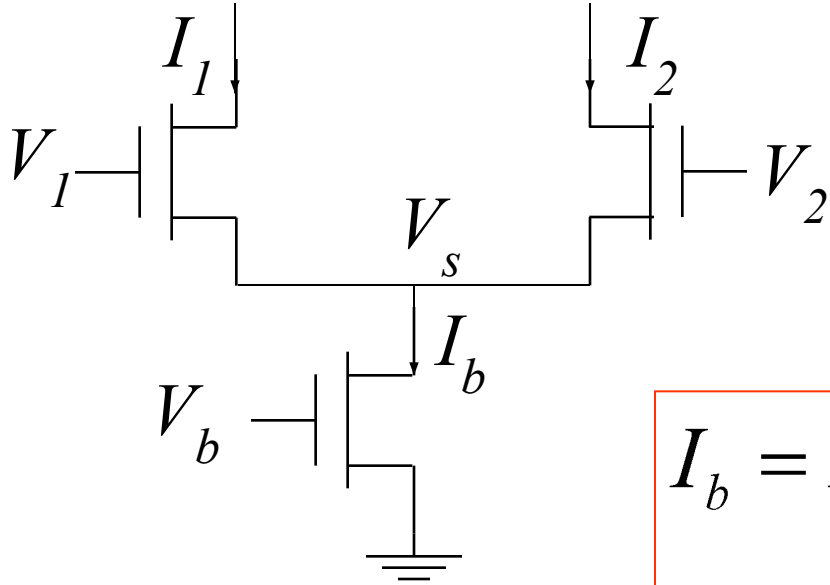
Outline:

- Differential pair
- Transconductance amplifier (and its g_m and A)
- Voltage amplifier
- Wide range transconductance amplifier

Differential Pair (I)



Differential Pair (II)



$$I_1 = I_0 e^{\kappa V_1 - V_s / U_T}$$
$$I_2 = I_0 e^{\kappa V_2 - V_s / U_T}$$

$$I_b = I_0 e^{\kappa V_b / U_T} = I_1 + I_2$$
$$= I_0 e^{-V_s / U_T} \left(e^{\kappa V_1 / U_T} + e^{\kappa V_2 / U_T} \right)$$

$$\Rightarrow e^{V_s / U_T} = \frac{e^{\kappa V_1 / U_T} + e^{\kappa V_2 / U_T}}{e^{\kappa V_b / U_T}}$$

Differential Pair (IV)

Output currents:

$$I_1 = I_b \frac{e^{\kappa V_1/U_T}}{e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T}} = \frac{I_b}{1 + e^{\kappa(V_2 - V_1)/U_T}}$$

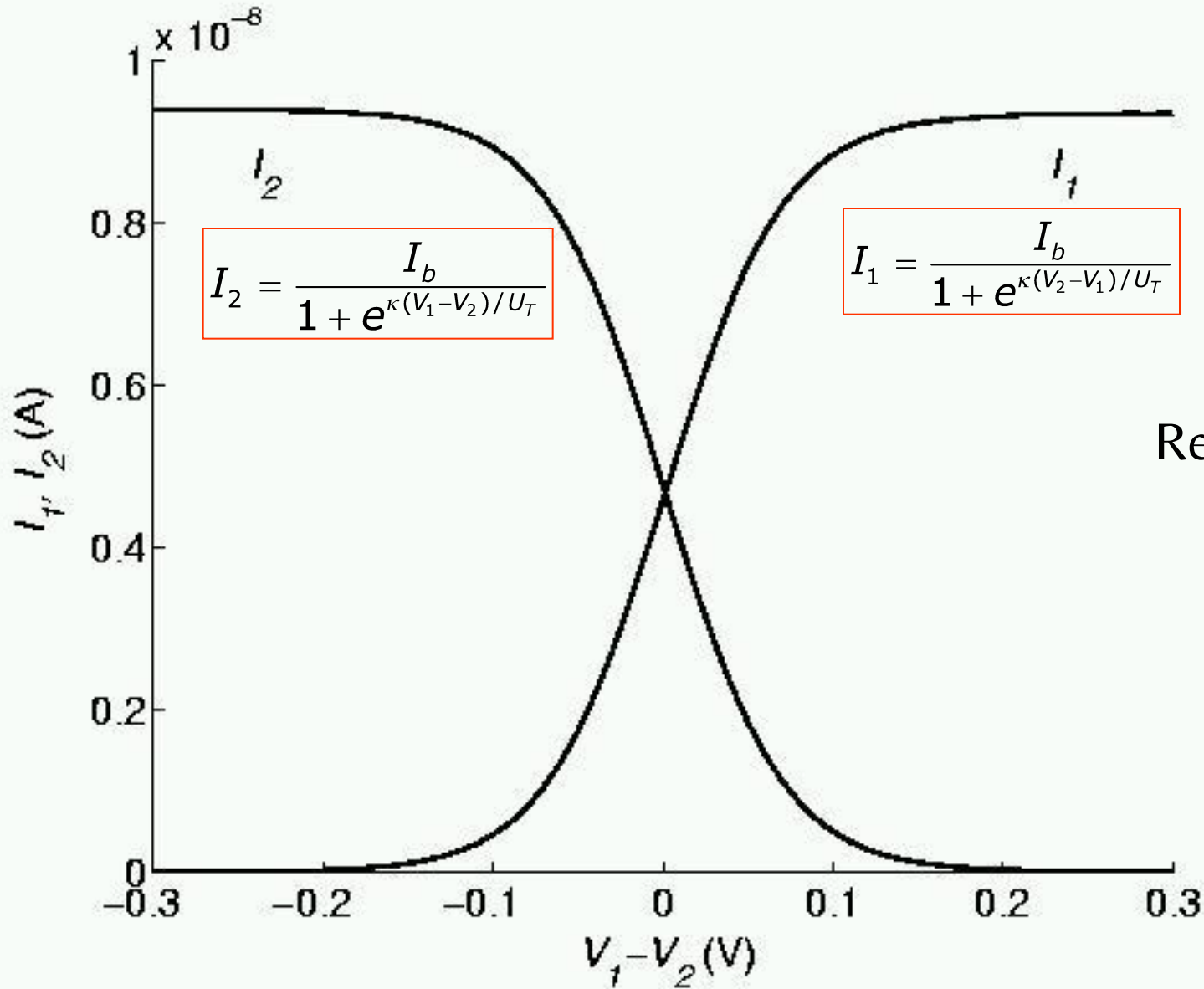
$$I_2 = \frac{I_b}{1 + e^{\kappa(V_1 - V_2)/U_T}}$$

Fermi function

Difference of output currents:

$$I_1 - I_2 = I_b \frac{e^{\kappa V_1/U_T} - e^{\kappa V_2/U_T}}{e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T}} = I_b \tanh \left(\kappa \frac{V_1 - V_2}{2U_T} \right)$$

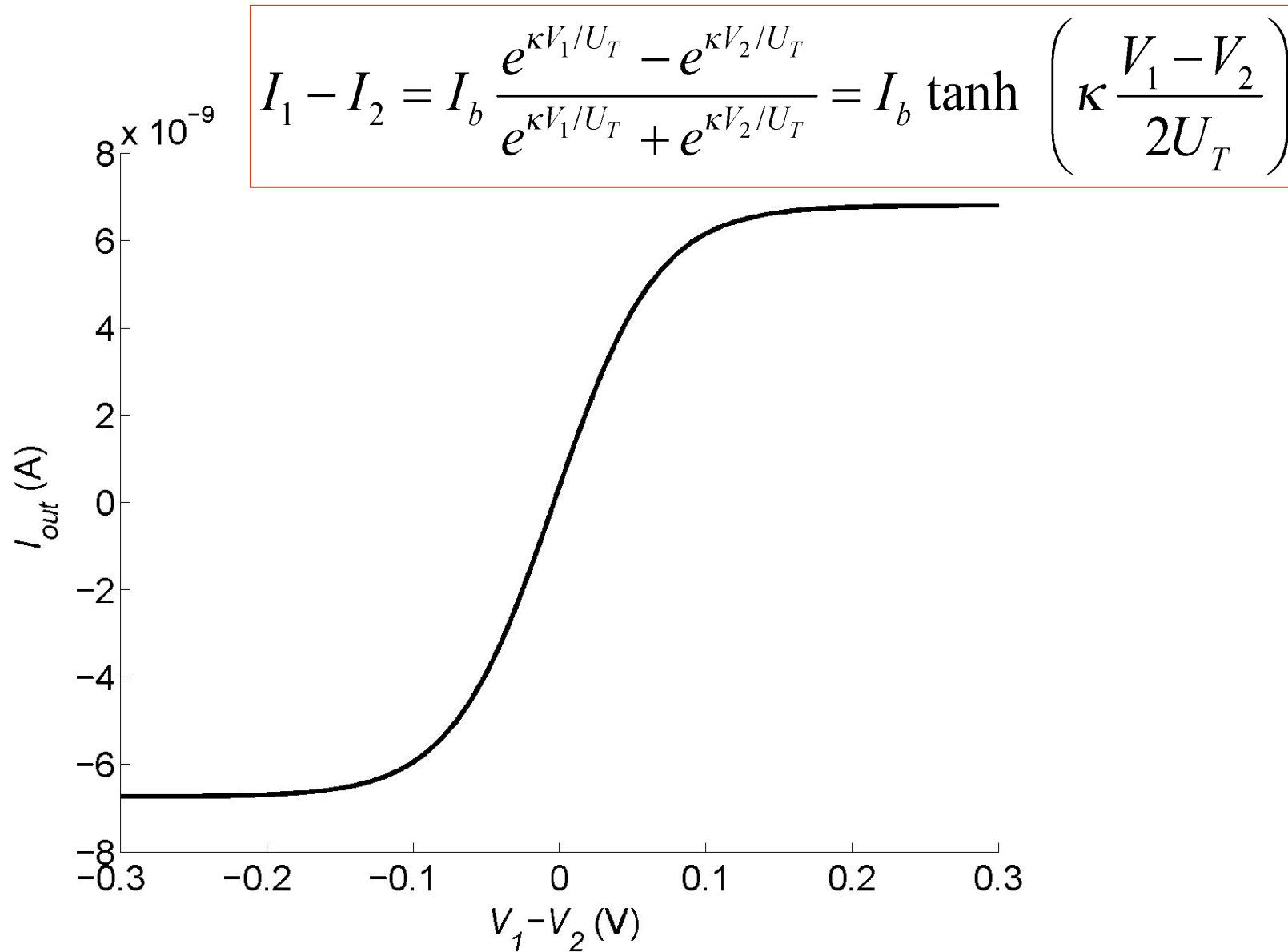
Differential Pair (IV)



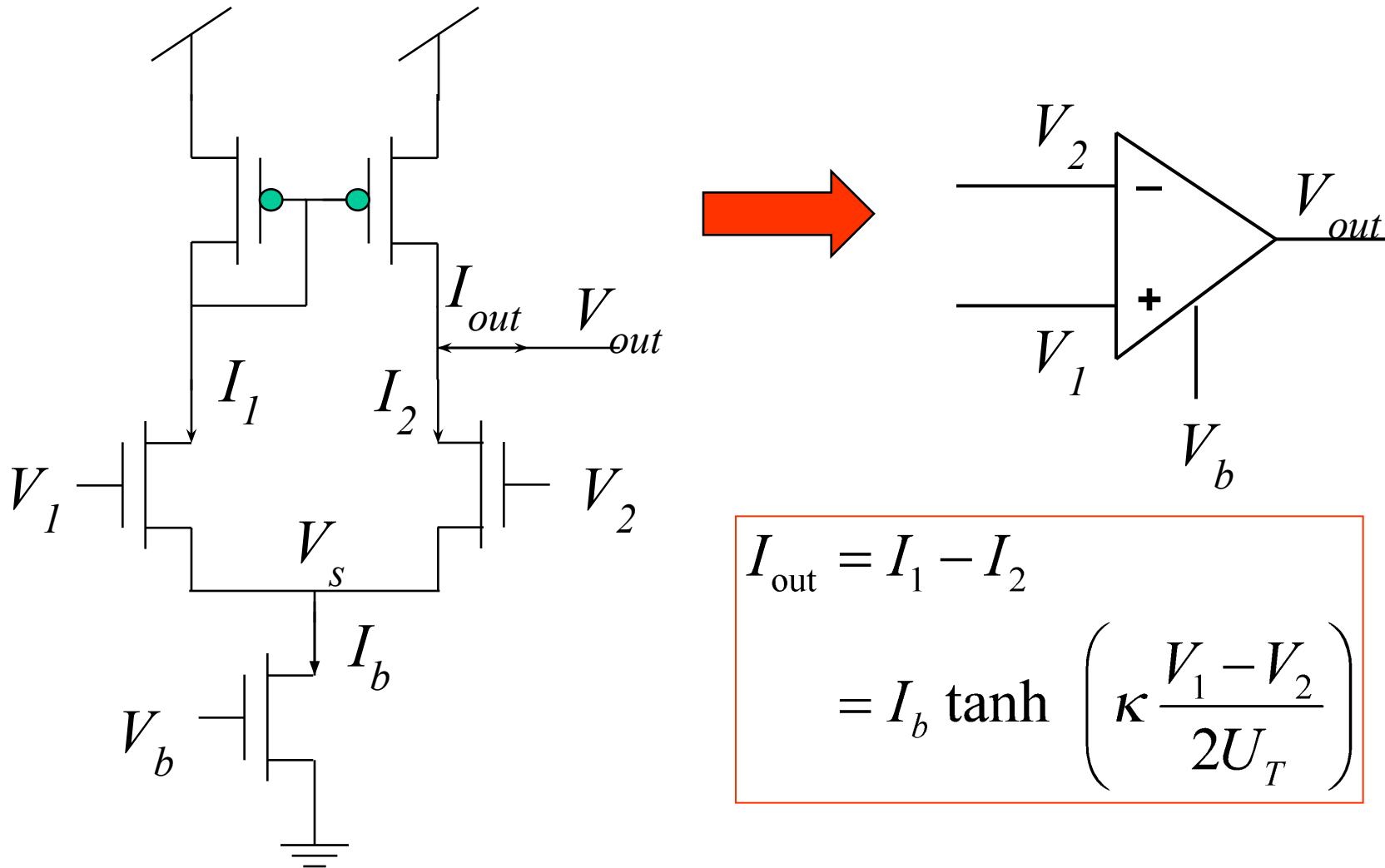
Remember sigmoid is

$$f(x) = \frac{1}{1 + e^{-\alpha x}}$$

Output Current vs Differential Input



Transconductance Amplifier



Transconductance Amplifier

In subthreshold:

For small differential voltages ($|V_1 - V_2|$ e.g. $< 200\text{mV}$),
the $\tanh(.)$ relationship is approximately linear :

$$I_{out} = I_b \tanh \left(\kappa \frac{V_1 - V_2}{2U_T} \right)$$

can be reduced to :

$$I_{out} \approx g_m (V_1 - V_2)$$

where

$$g_m \approx \frac{\kappa I_b}{2U_T}$$

Transconductance Amplifier

Output conductance is defined as :

$$g_d = -\frac{\partial I_{out}}{\partial V_{out}} \approx \frac{I_b}{V_E}$$

Transconductance in Strong Inversion

In above threshold, the output current of the diff pair is

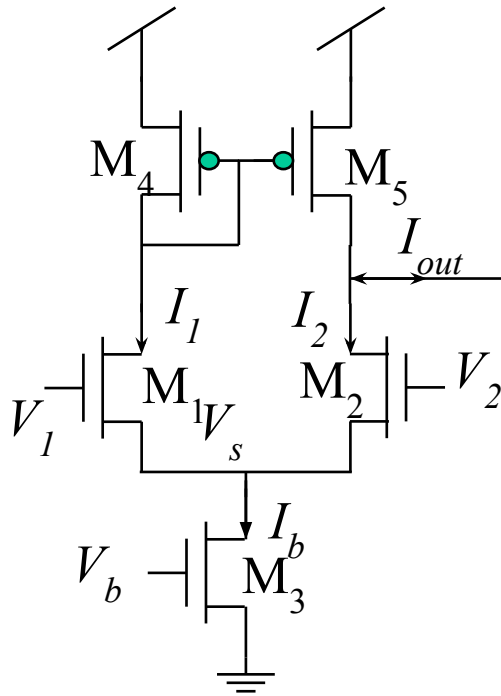
$$I_1 - I_2 = \frac{\beta}{2}(V_1 - V_2) \sqrt{\frac{4I_b}{\beta} - (V_1 - V_2)^2}$$

where $\beta = \mu C_{ox} \frac{W}{L}$.

For $|V_1 - V_2| = \sqrt{2I_b / \beta}$, the transconductance is given by

$$g_m = \sqrt{\beta I_b}$$

Assumptions for Deriving Tanh Output



To obtain tanh function, we assume M_1 to M_5 are in saturation:

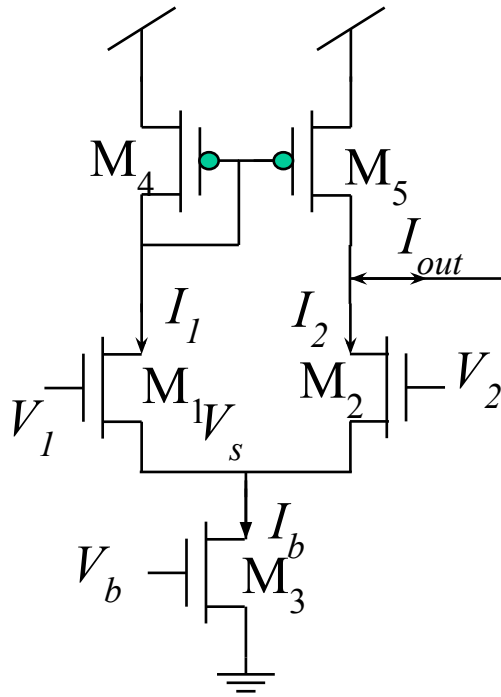
$$I_1 = I_0 e^{\kappa V_1 - V_s / U_T}$$

$$I_2 = I_0 e^{\kappa V_2 - V_s / U_T}$$

$$I_b = I_0 e^{\kappa V_b / U_T}$$

$$I_{out} = I_1 - I_2$$

Deriving Common Source (I)



Assume bias transistor is not in saturation,
Equation for current through M_3 is

$$I_b = I_0 e^{\kappa V_b / U_T} (1 - e^{-V_s / U_T})$$

Assuming M_1 and M_2 are in saturation:

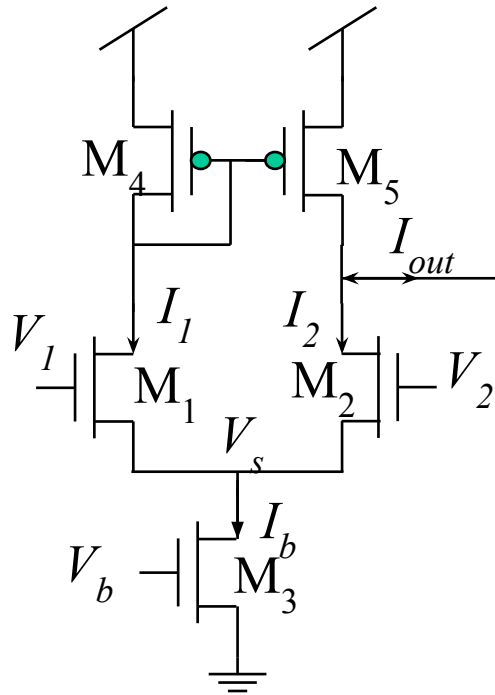
$$I_b = I_1 + I_2$$

$$e^{\kappa V_b / U_T} (1 - e^{-V_s / U_T}) = e^{(\kappa V_1 - V_s) / U_T} + e^{(\kappa V_2 - V_s) / U_T}$$

Solving for :

$$e^{-V_s / U_T} = \frac{e^{\kappa V_b / U_T}}{e^{\kappa V_b / U_T} + e^{\kappa V_1 / U_T} + e^{\kappa V_2 / U_T}} \quad (Eq.1)$$

Deriving Common Source (II)



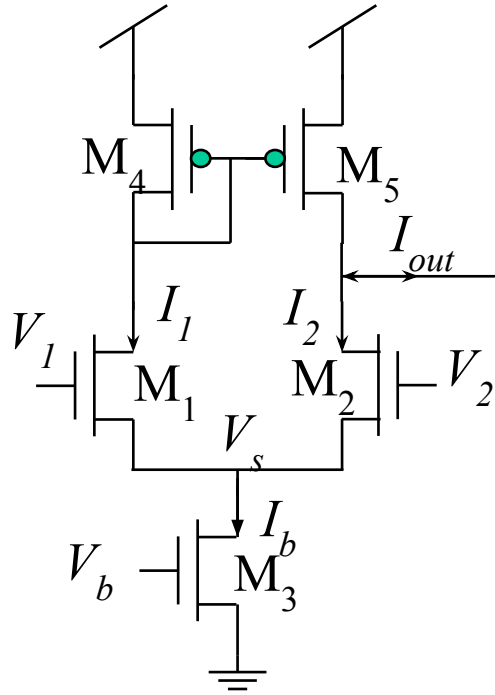
For the bias transistor to be in saturation,
 $V_s > 4U_T$ therefore $e^{-V_s/U_T} \ll 1$

$$e^{-V_s/U_T} = \frac{e^{\kappa V_b/U_T}}{e^{\kappa V_b/U_T} + e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T}} \ll 1$$

$$e^{\kappa V_b/U_T} \ll e^{\kappa V_b/U_T} + e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T}$$

$$\frac{e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T}}{e^{\kappa V_b/U_T}} \gg 1 \quad (Eq.2)$$

Deriving Common Source (III)



From

$$e^{-V_s/U_T} = \frac{e^{\kappa V_b/U_T}}{e^{\kappa V_b/U_T} + e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T}}$$

we can derive:

$$V_s = -\kappa V_b + U_T \ln(e^{\kappa V_b/U_T} + e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T})$$

Since $e^{\kappa V_1/U_T} + e^{\kappa V_2/U_T} \gg e^{\kappa V_b/U_T}$ and assuming that $|V_1 - V_2| > 4U_T$ we can simplify the $\ln(\cdot)$.

Deriving Common Source (IV)

The common node voltage V_s of the transconductance amplifier is:

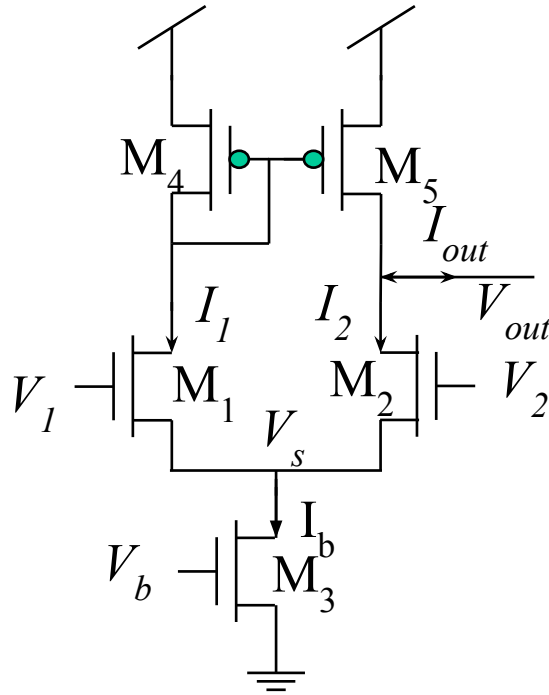
$$V_s = \kappa \left(\max(V_1, V_2) - V_b \right)$$

and the saturation condition for M_3 , $V_s > 4U_T$ is :

$$\max(V_1, V_2) > V_b + \frac{4U_T}{\kappa}$$

Output Transistors

The output transistors M_2 and M_5 should also be in saturation:



To keep M_5 in saturation : $V_{dd} - V_{out} > 4U_T$

To keep M_2 in saturation : $V_{out} - V_s > 4U_T$

which means

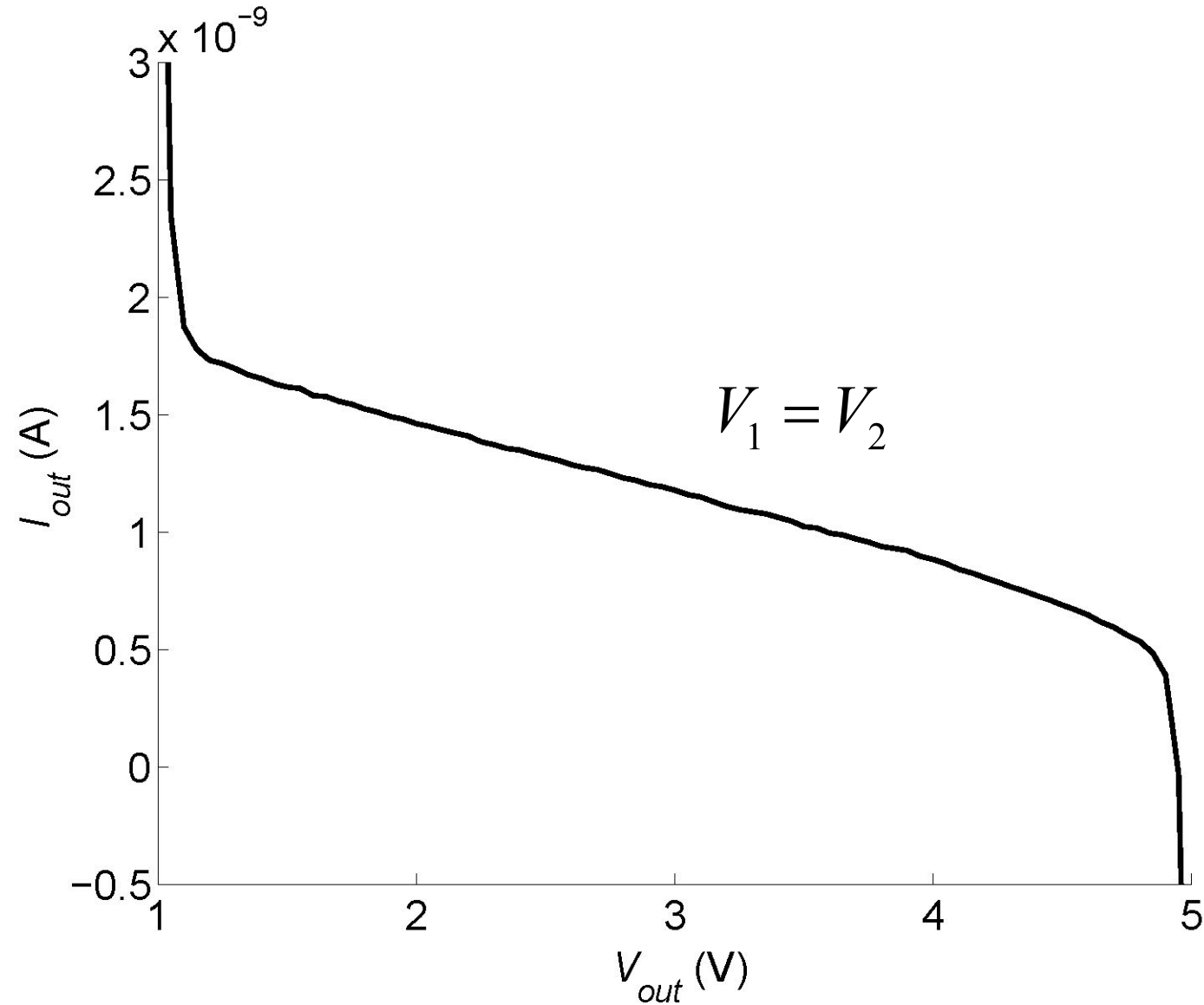
$$V_{out} > \kappa (\max(V_1, V_2) - V_b) + 4U_T \quad (Eq.3)$$

Therefore there is a V_{min} problem, i.e. minimum V_{out} depends on V_1 , V_2 and V_b .

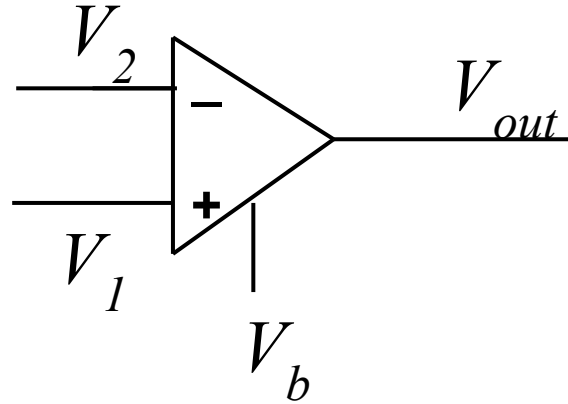
Final condition :

$$\kappa (\max(V_1, V_2) - V_b) + 4U_T < V_{out} < V_{dd} - 4U_T$$

Curves of I_{out} vs V_{out}



Voltage Amplifier



Transconductance amplifier can be used a differential-input voltage amplifier:

$$V_{out} = A(V_1 - V_2)$$

Transfer function :

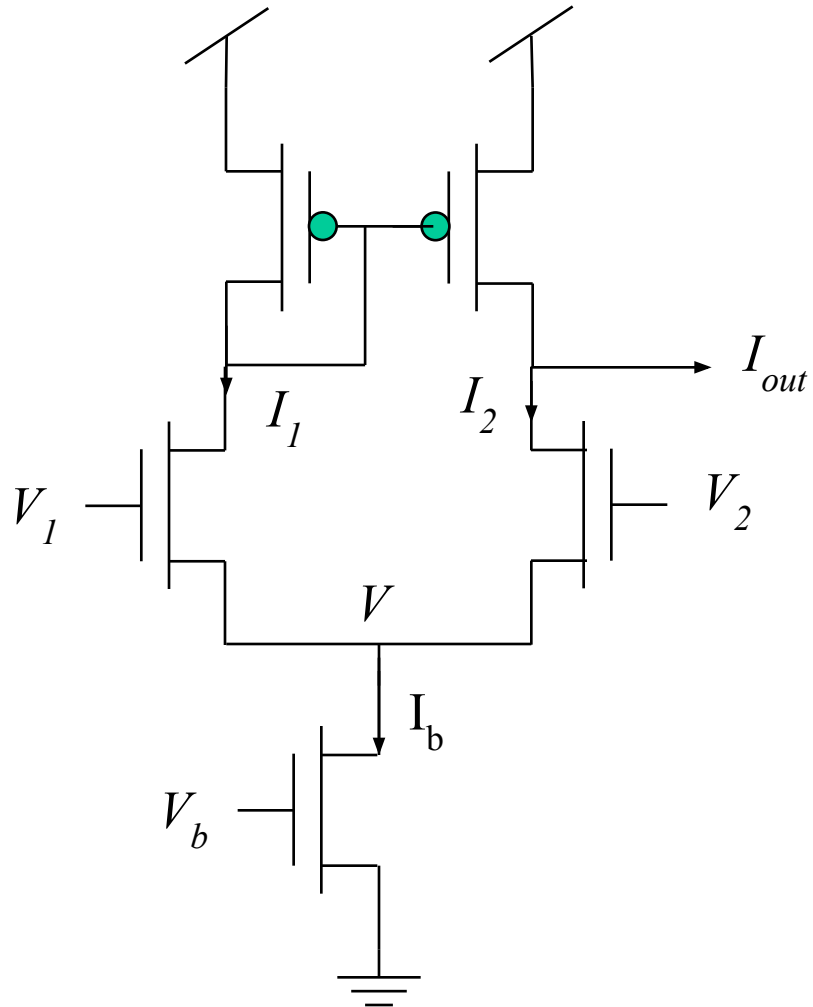
$$A = \frac{dV_{out}}{d(V_1 - V_2)} = \frac{dI_{out}}{d(V_1 - V_2)} \frac{dV_{out}}{dI_{out}} = \frac{g_m}{g_d}$$

$$\text{Subthreshold : } A \approx \frac{\kappa V_E}{2U_T}; \text{ Above threshold : } A \approx \sqrt{\frac{\beta}{I_b}} V_E$$

Voltage Amplifier

- The open-circuit voltage gain A increases with Early voltage, and therefore with the length of the output transistors.
- Typical subthreshold values are between 100 and 1000.
- Because of the large voltage gain and transistor mismatch effects, the amplifier is usually used in a negative-feedback configuration.
- In open-voltage mode, it is used mainly as a comparator. V_{out} is “high” when $V_1 > V_2$ and vice-versa.

Transconductance Amplifier



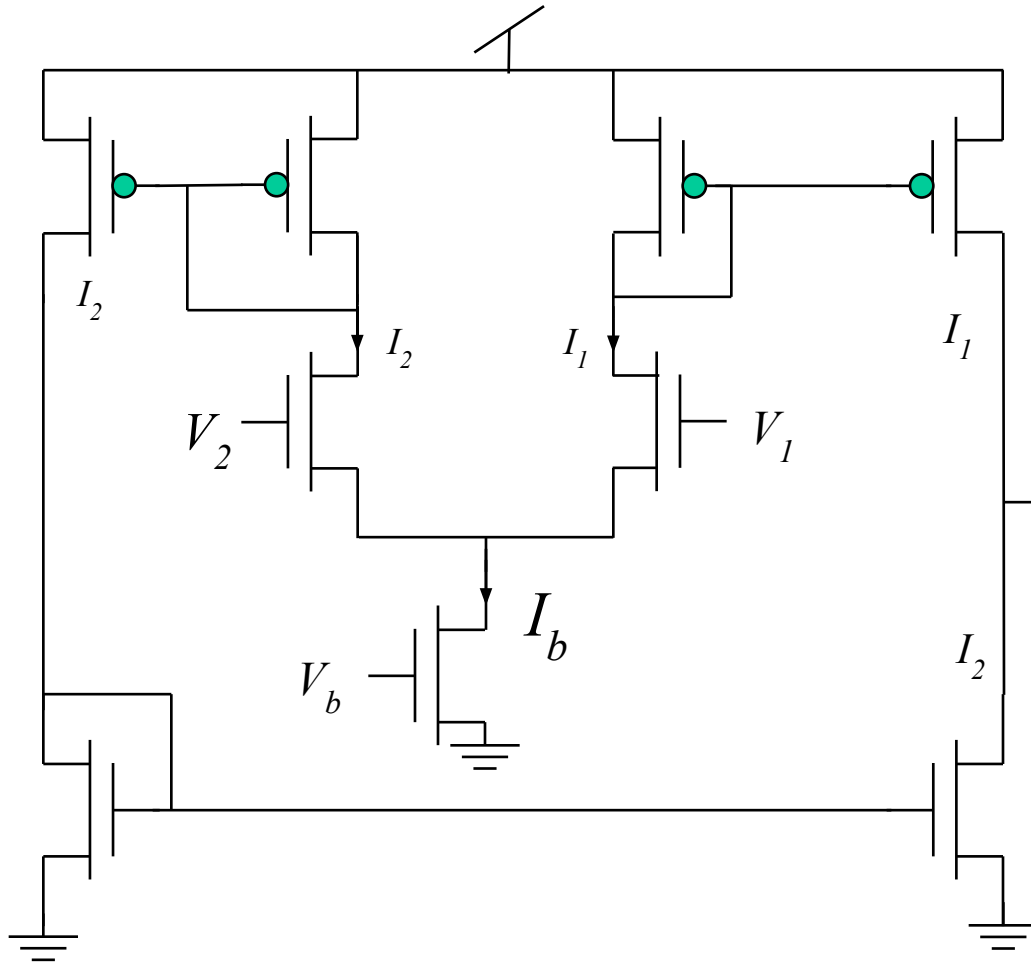
$$I_{out} = I_1 - I_2$$

$$= I_b \tanh \left(\kappa \frac{V_1 - V_2}{2U_T} \right)$$

Good: Simple, cheap

Bad: output voltage is restricted, voltage gain is limited

Wide-Output Range Transconductance Amplifier



$$I_{out} = I_b \tanh \left(\kappa \frac{V_1 - V_2}{2U_T} \right)$$

Good: output voltage is unrestricted, gain can be $>10^4$

Bad: larger, more mismatch

THE END

Next week: Linear systems